The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 April 2009.

METRIC

MIL-PRF-19500/741A
30 January 2009
SUPERSEDING
MIL-PRF-19500/741
14 March 2006

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, FIELD EFFECT, RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR DIE, N-CHANNEL AND P-CHANNEL, SILICON, VARIOUS TYPES, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel and P-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor die. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 through 8 herein.

1.3 Maximum ratings. See the applicable performance specification sheet from table I herein.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

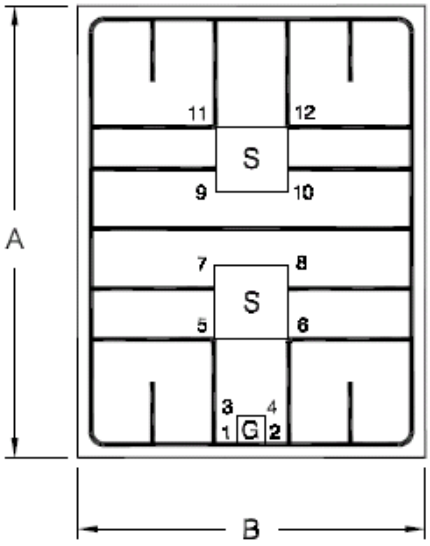


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**2N7467, 2N7478, 2N7468, 2N7470**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 8634.0 | 339.9 | ±200 µm (7.87 mils) |
| B | 6602.0 | 259.9 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541.00 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.30 | 0 | 21.193 | |
| 4 | 541.0 | 538.30 | 21.299 | 21.193 | |
| 5 | -424.1 | 2005.1 | -16.7 | 78.941 | |
| 6 | 969.0 | 2005.1 | 38.15 | 78.941 | |
| 7 | -424.1 | 3397.5 | -16.7 | 133.76 | |
| 8 | 969.0 | 3397.5 | 38.15 | 133.76 | |
| 9 | -424.1 | 4814.1 | -16.7 | 189.53 | |
| 10 | 965.1 | 4814.1 | 37.996 | 189.53 | |
| 11 | -424.1 | 6033.1 | -16.7 | 237.52 | |
| 12 | 965.1 | 6033.1 | 37.996 | 237.52 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 330 µm (13 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate(G)/source(S) pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 1. JANHC and JANKC A-version die dimensions, 2N7467, 2N7478, 2N7468, and 2N7470.

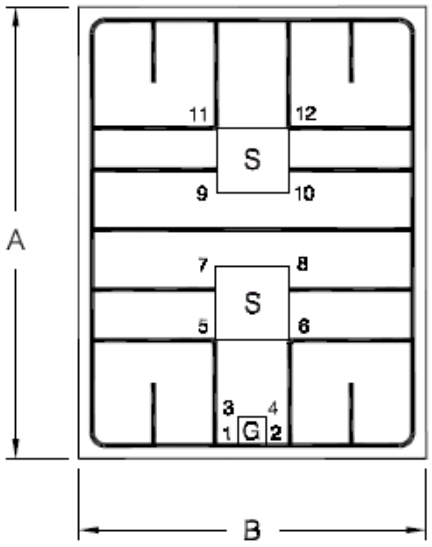


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**2N7469, 2N7471, 2N7472, 2N7475, 2N7473, 2N7476, 2N7474, 2N7477, 2N7549, 2N7550**

| Ltr. | Dimensions | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | Min | Max | Min | Max | |
| A | 9144.0 | | 360.0 | | ±200 µm (7.87 mils) |
| B | 6602.0 | | 259.9 | | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541.0 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.3 | 0 | 21.193 | |
| 4 | 541.0 | 538.3 | 21.299 | 21.193 | |
| 5 | -424.1 | 2199.6 | -16.7 | 86.598 | |
| 6 | 969.0 | 2199.6 | 38.150 | 86.598 | |
| 7 | -424.1 | 3592.0 | -16.7 | 141.42 | |
| 8 | 969.0 | 3592.0 | 38.150 | 141.42 | |
| 9 | -424.1 | 5013.9 | -16.7 | 197.40 | |
| 10 | 965.1 | 5013.9 | 37.996 | 197.40 | |
| 11 | -424.1 | 6232.9 | -16.7 | 245.39 | |
| 12 | 965.1 | 6232.9 | 37.996 | 245.39 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 330 µm (13 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate (G)/source (S) pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 2. JANHC and JANKC A-version die dimensions, 2N7469, 2N7471, 2N7472, 2N7475, 2N7473, 2N7476, 2N7474, 2N7477, 2N7549, 2N7550.

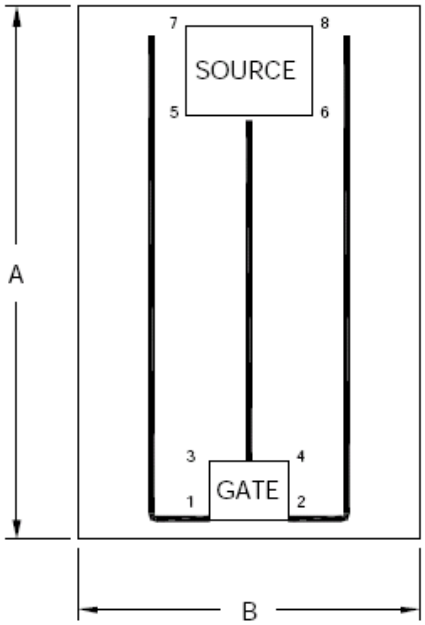


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**2N7479, 2N7482, 2N7491, 2N7494, 2N7480, 2N7483, 2N7492, 2N7495**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 509.9 | 0 | 20.075 | |
| 4 | 685.9 | 509.9 | 27.004 | 20.075 | |
| 5 | -206.1 | 3474.3 | -8.114 | 136.78 | |
| 6 | 871.9 | 3474.3 | 34.327 | 136.78 | |
| 7 | -206.1 | 4235.4 | -8.114 | 166.75 | |
| 8 | 871.9 | 4235.4 | 34.327 | 166.75 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 330 µm (13 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 3. JANHC and JANKC A-version die dimensions, 2N7479, 2N7482, 2N7491, 2N7494, 2N7480, 2N7483, 2N7492, 2N7495.

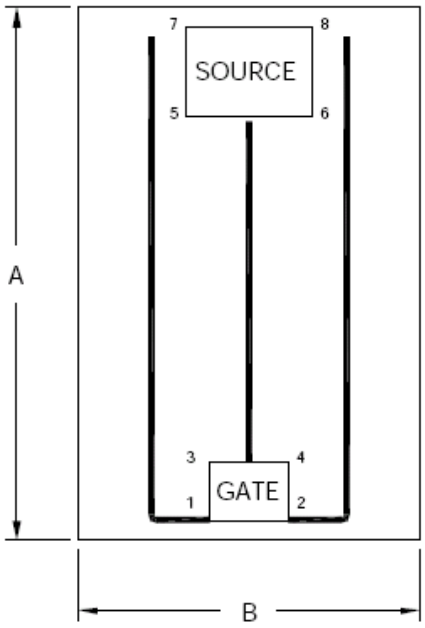


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**2N7481, 2N7484, 2N7493, 2N7496, 2N7545, 2N7547**

| Ltr | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 509.9 | 0 | 20.075 | |
| 4 | 685.9 | 509.9 | 27.004 | 20.075 | |
| 5 | -206.1 | 3398.5 | -8.114 | 133.8 | |
| 6 | 871.9 | 3398.5 | 34.327 | 133.8 | |
| 7 | -206.1 | 4155.6 | -8.114 | 163.61 | |
| 8 | 871.9 | 4155.6 | 34.327 | 163.61 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 330 µm (13 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 4. JANHC and JANKC A-version die dimensions, 2N7481, 2N7484, 2N7493, 2N7496, 2N7545, and 2N7547.

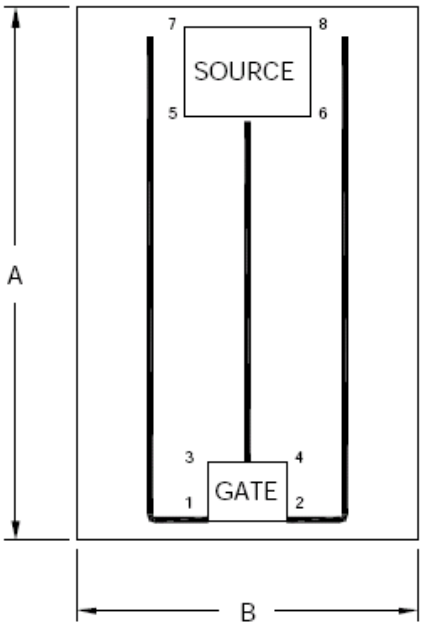


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**2N7485, 2N7488, 2N7497, 2N7500, 2N7486, 2N7489, 2N7498, 2N7501, 2N7487, 2N7490, 2N7499, 2N7502, 2N7546, 2N7548**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 508.0 | 0 | 20.000 | |
| 4 | 685.9 | 508.0 | 27.004 | 20.000 | |
| 5 | -206.1 | 3357.3 | -8.114 | 132.18 | |
| 6 | 891.9 | 3357.3 | 35.114 | 132.18 | |
| 7 | -206.1 | 4358.0 | -8.114 | 171.58 | |
| 8 | 891.9 | 4358.0 | 35.114 | 171.58 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 203 µm (8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 5. JANHC and JANKC A-version die dimensions, 2N7485, 2N7488, 2N7497, 2N7500, 2N7486, 2N7489, 2N7498, 2N7501, 2N7487, 2N7490, 2N7499, 2N7502, 2N7546, and 2N7548.

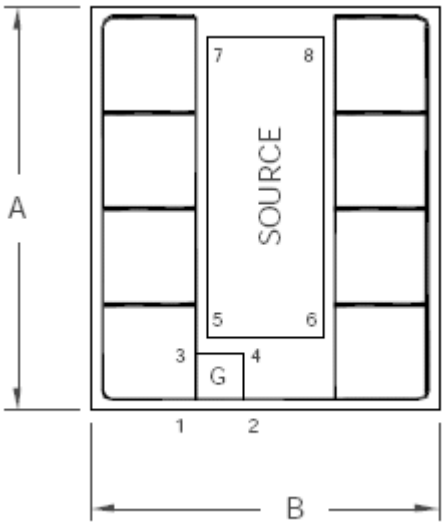


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**2N7519, 2N7520**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4610.8 | 181.5 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541 | 0 | 21.299 | 0 | |
| 3 | 0 | 537.6 | 0 | 21.165 | |
| 4 | 541 | 537.6 | 21.299 | 21.165 | |
| 5 | 72 | 929.9 | 2.835 | 36.610 | |
| 6 | 1170 | 929.9 | 46.063 | 36.610 | |
| 7 | 72 | 4017.9 | 2.835 | 158.185 | |
| 8 | 1170 | 4017.9 | 46.063 | 158.185 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 203 µm (8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate (G)/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 6. JANHC and JANKC A-version die dimensions, 2N7519 and 2N7520.

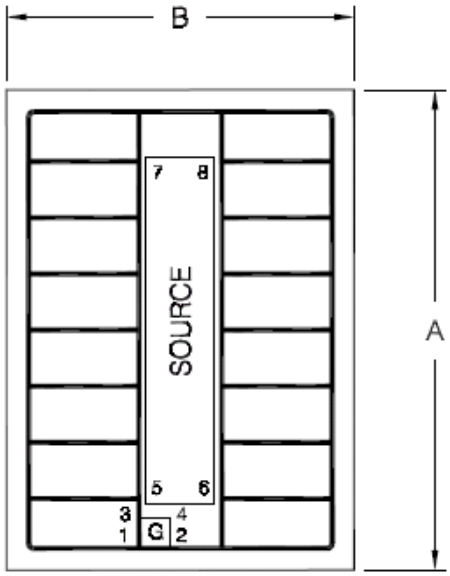


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**2N7523, 2N7524**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 9144 | 360.0 | ±200 µm (7.87 mils) |
| B | 6604 | 260.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.3 | 0 | 21.193 | |
| 4 | 541 | 538.3 | 21.299 | 21.193 | |
| 5 | 73.7 | 1023 | 2.902 | 40.276 | |
| 6 | 1528.8 | 1023 | 60.189 | 40.276 | |
| 7 | 73.7 | 7737 | 2.902 | 304.606 | |
| 8 | 1528.8 | 7737 | 60.189 | 304.606 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 203 µm (8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate (G)/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

FIGURE 7. JANHC and JANKC A-version die dimensions, 2N7523 and 2N7524.

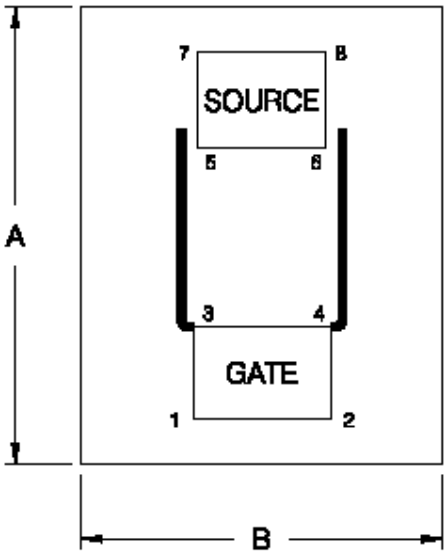


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**2N7503, 2N7506, 2N7518 N or P channel, 2N7521, 2N7525, 2N7522, 2N7526**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 2542.9 | 100.1 | ±200 µm (7.87 mils) |
| B | 2010.0 | 79.1 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 502.0 | 0 | 19.764 | |
| 4 | 685.9 | 502.0 | 27.004 | 19.764 | |
| 5 | -10.0 | 1560.5 | -0.394 | 61.437 | |
| 6 | 695.8 | 1560.5 | 27.394 | 61.437 | |
| 7 | -10.0 | 2099.9 | -0.394 | 82.673 | |
| 8 | 695.8 | 2099.9 | 27.394 | 82.673 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 330 µm (13 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.5 for ordering information.

* FIGURE 8. JANHC and JANKC A-version die dimensions, 2N7503, 2N7506, 2N7518-N or P channel, 2N7521, 2N7525, 2N7522, and 2N7526.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 through 8.

3.5 Marking. The individual die are not required to be marked. The die container (waffle pack) shall be marked in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.6).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Avoid use of plastic, rubber, or silk in MOS areas.

d. Maintain relative humidity above 50 percent if practical.

e. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any pad.

f. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.



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3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in the applicable specification sheet listed in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (element evaluation) (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection (element evaluation). Qualification inspection (element evaluation) shall be in accordance with appendix G of MIL-PRF-19500 and the applicable specification sheet from table I herein.

4.3 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix G of MIL-PRF-19500 and the applicable specification sheet from table I herein, except test current shall not exceed 20 A.. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTXV requirements.

4.4 Conformance inspection (group D). Conformance inspection (group D) shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and the applicable performance specification sheet from table I herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Applicable specification sheets.

| Type (same exact die in:) | Specification sheet and common part number | Reference data | | | Figure |
|---|---|---|---|---|---|
| | | Voltage (V dc) | Channel | Die size | |
| 2N7467 or 2N7478 | MIL-PRF-19500/697, 2N7478 | 30 | N | 6 | 1 |
| 2N7467 or 2N7478 | MIL-PRF-19500/683, 2N7467 | 30 | N | 6 | 1 |
| 2N7468 or 2N7470 | MIL-PRF-19500/698, 2N7470 | 60 | N | 6 | 1 |
| 2N7469 or 2N7471 | MIL-PRF-19500/698, 2N7471 | 100 | N | 6 | 2 |
| 2N7472 or 2N7475 | MIL-PRF-19500/685, 2N7475 | 130 | N | 6 | 2 |
| 2N7473 or 2N7476 | MIL-PRF-19500/685, 2N7476 | 200 | N | 6 | 2 |
| 2N7474 or 2N7477 | MIL-PRF-19500/685, 2N7477 | 250 | N | 6 | 2 |
| 2N7479 or 2N7482 or 2N7491 or 2N7494 | MIL-PRF-19500/702, 2N7482 | 30 | N | 3 | 3 |
| 2N7480 or 2N7483 or 2N7492 or 2N7495 | MIL-PRF-19500/702, 2N7483 | 60 | N | 3 | 3 |
| 2N7481 or 2N7484 or 2N7493 or 2N7496 | MIL-PRF-19500/702, 2N7484 | 100 | N | 3 | 4 |
| 2N7485 or 2N7488 or 2N7497 or 2N7500 | MIL-PRF-19500/705, 2N7488 | 130 | N | 3 | 5 |
| 2N7486 or 2N7489 or 2N7498 or 2N7501 | MIL-PRF-19500/705, 2N7489 | 200 | N | 3 | 5 |
| 2N7487 or 2N7490 or 2N7499 or 2N7502 | MIL-PRF-19500/705, 2N7490 | 250 | N | 3 | 5 |
| * 2N7503 or 2N7518-Nch or 2N7521 or 2N7525 | MIL-PRF-19500/740, 2N7525 | 100 | N | 1 | 8 |
| 2N7523 | MIL-PRF-19500/733, 2N7523 | -30 | P | 6 | 7 |
| 2N7524 | MIL-PRF-19500/733, 2N7524 | -60 | P | 6 | 7 |
| 2N7550 | MIL-PRF-19500/713, 2N7550 | -100 | P | 6 | 2 |
| 2N7549 | MIL-PRF-19500/713, 2N7549 | -200 | P | 6 | 2 |
| 2N7519 | MIL-PRF-19500/732, 2N7519 | -30 | P | 3 | 6 |
| 2N7520 | MIL-PRF-19500/732, 2N7520 | -60 | P | 3 | 6 |
| 2N7545 or 2N7547 | MIL-PRF-19500/712, 2N7547 | -100 | P | 3 | 4 |
| 2N7546 or 2N7548 | MIL-PRF-19500/712, 2N7548 | -200 | P | 3 | 5 |
| * 2N7506 or 2N7518-Pch or 2N7522 or 2N7526 | MIL-PRF-19500/740, 2N7526 | -100 | P | 1 | 8 |



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Product assurance level and type designator.

d. Specify the JANHC or JANKC letter version (see 6.5 and figures 1 through 8 herein).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .



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* 6.4 Cross reference list. The following information shows the generic P/N and its' associated military P/N (without the JAN or RHA prefix). Multiple military part number indicates that the same die type is used on more than one specification sheet.

| For any military die-packaged basic P/N | Common military die P/N (1) | Generic die P/N (2) |
|---|---|---|
| 2N7467 or 2N7478 | JANHCAX or JANKCAX2N7478 | IRHC5xZ60 |
| 2N7468 or 2N7470 | JANHCAX or JANKCAX2N7470 | IRHC5x064 |
| 2N7469 or 2N7471 | JANHCAX or JANKCAX2N7471 | IRHC5x160 |
| 2N7472 or 2N7475 | JANHCAX or JANKCAX2N7475 | IRHC5x163SE |
| 2N7473 or 2N7476 | JANHCAX or JANKCAX2N7476 | IRHC5x260SE |
| 2N7474 or 2N7477 | JANHCAX or JANKCAX2N7477 | IRHC5x264SE |
| 2N7479 or 2N7482 or 2N7491 or 2N7494 | JANHCAX or JANKCAX2N7482 | IRHC5xZ30 |
| 2N7480 or 2N7483 or 2N7492 or 2N7495 | JANHCAX or JANKCAX2N7483 | IRHC5x034 |
| 2N7481 or 2N7484 or 2N7493 or 2N7496 | JANHCAX or JANKCAX2N7484 | IRHC5x130 |
| 2N7485 or 2N7488 or 2N7497 or 2N7500 | JANHCAX or JANKCAX2N7488 | IRHC5x133SE |
| 2N7486 or 2N7489 or 2N7498 or 2N7501 | JANHCAX or JANKCAX2N7489 | IRHC5x230SE |
| 2N7487 or 2N7490 or 2N7499 or 2N7502 | JANHCAX or JANKCAX2N7490 | IRHC5x234SE |
| 2N7521 or 2N7525 or 2N7503 or 2N7518 N-channel | JANHCAX or JANKCAX2N7525 | IRHC5x110 |
| 2N7523 | JANHCAX or JANKCAX2N7523 | IRHC59xZ60 |
| 2N7524 | JANHCAX or JANKCAX2N7524 | IRHC59x064 |
| 2N7550 | JANHCAX or JANKCAX2N7550 | IRHC59x160 |
| 2N7549 | JANHCAX or JANKCAX2N7549 | IRHC59x260 |
| 2N7519 | JANHCAX or JANKCAX2N7519 | IRHC59xZ30 |
| 2N7520 | JANHCAX or JANKCAX2N7520 | IRHC59x034 |
| 2N7545 or 2N7547 | JANHCAX or JANKCAX2N7547 | IRHC59x130 |
| 2N7546 or 2N7548 | JANHCAX or JANKCAX2N7548 | IRHC59x230 |
| 2N7522 or 2N7526 or 2N7506 or 2N7518 P-channel | JANHCAX or JANKCAX2N7526 | IRHC59x110 |

(1) These common die part numbers should be used for the same exact die used in their respective die-packaged part numbers. Replace X with letter indicating qualified rad-hardness (RHA designator) as follows:

R = 100 K Rad (Si) (equivalent to manufacturer designator 7).
F = 300 K Rad (Si) (equivalent to manufacturer designator 3).
G = 500 K Rad (Si) (equivalent to manufacturer designator 4).
H = 1000 K Rad (Si) (equivalent to manufacturer designator 8).

(2) Replace x with number indicating qualified rad-hardness as follows:

7 = 100 K Rad (Si) (equivalent to military RHA designator R).
3 = 300 K Rad (Si) (equivalent to military RHA designator F).
4 = 500 K Rad (Si) (equivalent to military RHA designator G).
8 = 1000 K Rad (Si) (equivalent to military RHA designator H).



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* 6.5 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N7261) will be identified on the QML-19500.

| JANC ordering information | |
|---|---|
| For die in type | Manufacturer CAGE 59993 |
| | Common JANC die part numbers for ordering |
| 2N7467 or 2N7478 | JANHCAR2N7478, JANKCAR2N7478, JANHCAF2N7478, JANKCAF2N7478<br>JANHCAG2N7478, JANKCAG2N7478, JANHCAH2N7478, JANKCAH2N7478 |
| 2N7468 or 2N7470 | JANHCAR2N72470, JANKCAR2N7470, JANHCAF2N7470, JANKCAF2N7470<br>JANHCAG2N72470, JANKCAG2N7470, JANHCAH2N7470, JANKCAH2N7470 |
| 2N7469 or 2N7471 | JANHCAR2N7471, JANKCAR2N7471, JANHCAF2N7471, JANKCAF2N7471<br>JANHCAG2N7471, JANKCAG2N7471, JANHCAH2N7471, JANKCAH2N7471 |
| 2N7472 or 2N7475 | JANHCAR2N7475, JANKCAR2N7475, JANHCAF2N7475, JANKCAF2N7475 |
| 2N7473 or 2N7476 | JANHCAR2N7476, JANKCAR2N7476, JANHCAF2N7476, JANKCAF2N7476 |
| 2N7474 or 2N7477 | JANHCAR2N7477, JANKCAR2N7477, JANHCAF2N7477, JANKCAF2N7477 |
| 2N7479 or 2N7482 or 2N7491 or 2N7494 | JANHCAR2N7482, JANKCAR2N7482, JANHCAF2N7482, JANKCAF2N7482<br>JANHCAG2N7482, JANKCAG2N7482, JANHCAH2N7482, JANKCAH2N7482 |
| 2N7480 or 2N7483 or 2N7492 or 2N7495 | JANHCAR2N7483, JANKCAR2N7483, JANHCAF2N7483, JANKCAF2N7483<br>JANHCAG2N7483, JANKCAG2N7483, JANHCAH2N7483, JANKCAH2N7483 |
| 2N7481 or 2N7484 or 2N7493 or 2N7496 | JANHCAR2N7484, JANKCAR2N7484, JANHCAF2N7484, JANKCAF2N7484<br>JANHCAG2N7484, JANKCAG2N7484, JANHCAH2N7484, JANKCAH2N7484 |
| 2N7485 or 2N7488 or 2N7497 or 2N7500 | JANHCAR2N7488, JANKCAR2N7488, JANHCAF2N7488, JANKCAF2N7488 |
| 2N7486 or 2N7489 or 2N7498 or 2N7501 | JANHCAR2N7489, JANKCAR2N7489, JANHCAF2N7489, JANKCAF2N7489 |
| 2N7487 or 2N7490 or 2N7499 or 2N7502 | JANHCAR2N7490, JANKCAR2N7490, JANHCAF2N7490, JANKCAF2N7490 |
| 2N7503 or 2N7518 N-channel or 2N7521 or 2N7525 | JANHCAR2N7525, JANKCAR2N7525, JANHCAF2N7525, JANKCAF2N7525<br>JANHCAG2N7525, JANKCAG2N7525, JANHCAH2N7525, JANKCAH2N7525 |
| 2N7523 | JANHCAR2N7523, JANKCAR2N7523, JANHCAF2N7523, JANKCAF2N7523 |
| 2N7524 | JANHCAR2N7524, JANKCAR2N7524, JANHCAF2N7524, JANKCAF2N7524 |
| 2N7550 | JANHCAR2N7550, JANKCAR2N7550, JANHCAF2N7550, JANKCAF2N7550 |
| 2N7549 | JANHCAR2N7549, JANKCAR2N7549, JANHCAF2N7549, JANKCAF2N7549 |
| 2N7519 | JANHCAR2N7519, JANKCAR2N7519, JANHCAF2N7519, JANKCAF2N7519 |
| 2N7520 | JANHCAR2N7520, JANKCAR2N7520, JANHCAF2N7520, JANKCAF2N7520 |
| 2N7545 or 2N7547 | JANHCAR2N7547, JANKCAR2N7547, JANHCAF2N7547, JANKCAF2N7547 |
| 2N7546 or 2N7548 | JANHCAR2N7548, JANKCAR2N7548, JANHCAF2N7548, JANKCAF2N7548 |
| 2N7506 or 2N7518 P-channel or 2N7522 or 2N7526 | JANHCAR2N7526, JANKCAR2N7526, JANHCAF2N7526, JANKCAF2N7526 |



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